Cyber Threat Intel

Multimodal Retrieval-Augmented Generation for Threat Intelligence Synthesis

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***Abstract*—Modern Security Operations Centers (SOC) face critical bottlenecks in consuming and operationalizing Cyber Threat Intelligence (CTI). High-fidelity advisories from vendors are predominantly delivered as unstructured PDF documents, necessitating hours of manual parsing to extract behavioral tactics and explicit Indicators of Compromise (IOCs). This preliminary report documents the implementation and early evaluation of Cyber Threat Intel, an AI-powered analytical dashboard utilizing a Hybrid Retrieval-Augmented Generation (RAG) architecture. By fusing dense semantic vectorization (FAISS) with sparse lexical retrieval (BM25), the system deterministically queries private threat data while preserving complex tabular structures. Furthermore, an automated background Regular Expression engine actively extracts actionable network artifacts for immediate defensive deployment. Early evaluation metrics demonstrate high fidelity in exact-string retrieval and a 100% success rate in parametric confinement (anti-hallucination) guardrails, though cloud infrastructure API quotas present notable scaling bottlenecks.**

***Index Terms*—Retrieval-Augmented Generation, Cyber Threat Intelligence, FAISS, BM25, Large Language Models, Generative AI, Indicators of Compromise**

I. Introduction and Problem Context

The contemporary cyber threat landscape is characterized by the rapid evolution of Advanced Persistent Threats (APTs) and Ransomware-as-a-Service (RaaS) cartels. To combat these threats, organizations rely on Cyber Threat Intelligence (CTI) published by entities such as the Cybersecurity and Infrastructure Security Agency (CISA), Mandiant, and CrowdStrike. However, the operationalization of this intelligence is severely hindered by its format: unstructured, human-readable Portable Document Format (PDF) files.

Tier-3 SOC Analysts and Threat Hunters are tasked with translating 40-page technical narratives into actionable firewall patches and detection logic. Manually extracting specific Tactics, Techniques, and Procedures (TTPs) alongside granular Indicators of Compromise (IOCs)—such as IPv4 addresses, SHA-256 file hashes, and Common Vulnerability and Exposure (CVE) identifiers—is a highly inefficient process that drastically increases the mean time to respond (MTTR) to active network intrusions.

The objective of the Cyber Threat Intel project is to resolve this unstructured data problem by engineering an AI-powered Threat Intelligence Dashboard. Since the foundational project proposal, the system has successfully transitioned from theory to a functional prototype. The current implementation successfully ingests dense PDFs using advanced Markdown extraction to preserve tabular data, embeds this data into a dual-retrieval vector space, and serves analyst queries via a Streamlit interface.

This report details the architecture, the algorithmic implementation, the interface prototype, early inference metrics, and the enterprise-level cloud infrastructure challenges encountered during development.

II. Related Work

The intersection of Artificial Intelligence and Cybersecurity has historically focused on predictive modeling, such as training Convolutional Neural Networks (CNNs) on malware binaries or utilizing Random Forests for network anomaly detection. However, the advent of Large Language Models (LLMs) has shifted focus toward intelligence synthesis.

A. Standard Large Language Models in Cybersecurity

General-purpose LLMs exhibit significant limitations in high-stakes security environments. Models rely on pre-trained parametric memory, which has a static knowledge cutoff date. Consequently, they are often unaware of zero-day vulnerabilities or newly established threat actor infrastructure. Furthermore, standard LLMs are highly susceptible to "hallucinations"—the generation of syntactically correct but factually false information. In a SOC environment, hallucinating a malicious IP address can result in self-inflicted Denial of Service (DoS) if an analyst applies the false indicator to an enterprise firewall.

B. Retrieval-Augmented Generation (RAG)

To mitigate hallucinations, Lewis et al. introduced Retrieval-Augmented Generation (RAG) [1]. RAG decouples the knowledge base from the model's parametric memory by retrieving relevant context from an external database and injecting it directly into the generative prompt. While standard RAG architectures utilize dense vector embeddings to perform semantic search, recent literature [3] indicates that semantic search struggles with exact alphanumeric string matching, a critical requirement for retrieving cryptographic hashes and CVE identifiers. Cyber Threat Intel builds upon this foundation by implementing a hybrid retrieval mechanism to ensure both semantic comprehension and lexical exactitude.

C. Traditional CTI Parsing Approaches

Historically, SOCs have attempted to automate CTI ingestion using strict Regular Expressions (Regex) or early Named

Entity Recognition (NER) models (such as SpaCy). While Regex is highly effective for extracting IP addresses, it entirely fails to capture the behavioral context surrounding the IP address (e.g., distinguishing between an attacker's Command and Control server versus a legitimate victim IP mentioned in the same report). Cyber Threat Intel bridges this gap by combining generative contextual synthesis with rigid Regex post-processing.

III. Theoretical Foundations

The core innovation of the Cyber Threat Intel architecture lies in its mathematical approach to information retrieval, combining two fundamentally different search paradigms.

A. Sparse Lexical Retrieval (BM25)

Best Matching 25 (BM25) is a probabilistic retrieval framework based on Term Frequency-Inverse Document Frequency (TF-IDF). It excels at finding exact keyword matches. Given a query Q containing keywords $q_1, q_2, ..., q_n$, the BM25 score of a document D is calculated as:

$$\text{Score}(D,Q) = \sum_{i=1}^{n} \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)} \tag{1}$$

Where $f(q_i, D)$ is the term frequency of q_i in document D, $|D|$ is the length of the document, avgdl is the average document length in the corpus, and k_1 and b are free parameters (typically $k_1 \in [1.2, 2.0]$ and $b = 0.75$). The Inverse Document Frequency (IDF) penalizes common words and rewards rare, highly specific strings (such as a SHA-256 hash).

B. Dense Semantic Retrieval and FAISS HNSW

To comprehend abstract concepts (e.g., "lateral movement"), the system employs dense vector embeddings. Text chunks are mapped into a high-dimensional continuous vector space using Google's `gemini-embedding-001` model. The similarity between a query vector $\vec{q}$ and a document vector $\vec{d}$ is computed using Cosine Similarity:

$$\text{Cosine}(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{\|\vec{q}\| \|\vec{d}\|} = \frac{\sum_{i=1}^{n} q_i d_i}{\sqrt{\sum_{i=1}^{n} q_i^2} \sqrt{\sum_{i=1}^{n} d_i^2}} \tag{2}$$

To search thousands of 3,072-dimensional vectors in milliseconds, computing exact cosine distances for every pair is computationally prohibitive. Therefore, Cyber Threat Intel utilizes Facebook AI Similarity Search (FAISS) employing Hierarchical Navigable Small World (HNSW) graphs. HNSW constructs a multi-layered graph where the top layers contain long-range links for fast, macroscopic routing, and the bottom layers contain dense clusters for fine-grained nearest-neighbor calculations, achieving logarithmic search complexity $\mathcal{O}(\log(N))$.

C. Reciprocal Rank Fusion (RRF)

Because BM25 and FAISS yield fundamentally incomparable scoring metrics, they cannot be natively added together. Cyber Threat Intel employs Reciprocal Rank Fusion (RRF) to normalize and combine the results. The RRF score for a document $d \in D$ is defined as:

$$RRF(d) = \sum_{r \in R} \frac{1}{k + rank_r(d)} \tag{3}$$

Where R is the set of retrievers, $rank_r(d)$ is the position of document d in retriever r's sorted results, and k is a smoothing constant (set to 60) to prevent outlier ranks from dominating the ensemble.

IV. System Architecture and Pipeline

The system data flow is divided into three distinct operational phases: Ingestion, Hybrid Retrieval, and Generative Inference.

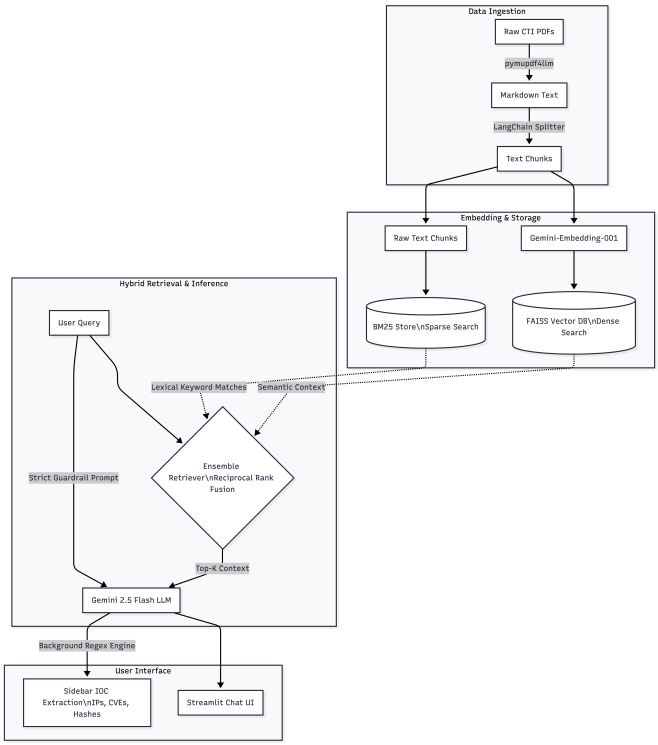

Fig. 1. Cyber Threat Intel Hybrid Data Flow Architecture.

A. Phase 1: Data Ingestion and Preprocessing

The foundation of any RAG architecture is the quality of its underlying text extraction. Standard PDF parsing libraries read documents sequentially. In cybersecurity reports, critical IOCs are almost exclusively stored in multi-column tables. Sequential parsing corrupts these tables, merging unrelated IP addresses.

To mitigate this, the preprocessing pipeline utilizes `pymupdf4llm` to extract the PDFs directly into raw Markdown format. This spatial-aware extraction guarantees that

tabular boundaries are preserved. The text is partitioned using LangChain's recursive character text splitter. To ensure complex behavioral narratives are not severed, the algorithm utilizes a `chunk_size` of 1,000 characters with a `chunk_overlap` of 200 characters.

B. Phase 2: Embedding and Storage

Text chunks are embedded using `gemini-embedding-001` into 3,072-dimensional vectors and stored locally in the FAISS index. Simultaneously, raw text chunks are indexed into the BM25 store.

C. Phase 3: Generation and Interface

When an analyst submits a query, it is executed against both FAISS and BM25 databases simultaneously. The results are mathematically merged via RRF, and the combined "Top-K" context is injected into the prompt of the `gemini-2.5-flash` LLM.

V. Implementation & Engineering Solutions

The system was engineered in Python, leveraging a modern, modular stack to orchestrate the retrieval and generation phases.

A. Infrastructure and Batching Algorithm

Due to the massive data density of enterprise threat reports (yielding over 4,000 distinct vector chunks for 75 PDFs), local ingestion initially triggered HTTP 429 `RESOURCE_EXHAUSTED` exceptions. Google Cloud's free-tier API imposes strict Requests-Per-Minute (RPM) and Daily Quota limits.

To ensure reproducibility and pipeline stability, a custom batch-processing mechanism was engineered. Algorithm 1 outlines the exact throttling logic implemented to bypass transient network rejections and respect API constraints.

Algorithm 1 Rate-Limited Vector Batch Ingestion

Require: $AllChunks$ (List of Document objects)
Require: $BATCH_SIZE \leftarrow 50$
Require: $SLEEP_TIME \leftarrow 30$ seconds
$VectorStore \leftarrow$ Null
for $i = 0$ to Length($AllChunks$) step $BATCH_SIZE$ **do**
  $Batch \leftarrow AllChunks[i$ to $i + BATCH_SIZE]$
  if $VectorStore$ is Null **then**
    $VectorStore \leftarrow$ FAISS.from_documents($Batch$)
  else
    $VectorStore$.add_documents($Batch$)
  end if
  if $i + BATCH_SIZE <$ Length($AllChunks$) **then**
    Print("Sleeping to respect API limits...")
    Time.sleep($SLEEP_TIME$)
  end if
end for
$VectorStore$.save_local("$faiss_index$")

B. Parametric Confinement via Prompt Engineering

To ensure the LLM strictly adheres to the retrieved context and refuses to hallucinate external knowledge, a rigid system prompt was engineered. The generative core receives instructions structured to enforce a zero-trust synthesis protocol. If the mathematical vector search does not return chunks containing the answer, the LLM is explicitly commanded to output "Insufficient data," overriding its natural conversational tendencies. The `gemini-2.5-flash` model is explicitly configured with `temperature = 0.1` to enforce this determinism.

VI. Interface Prototype and Usability

The user interface is deployed locally utilizing the Streamlit framework, designed to bridge the gap between complex vector mathematics and high-velocity SOC operations.

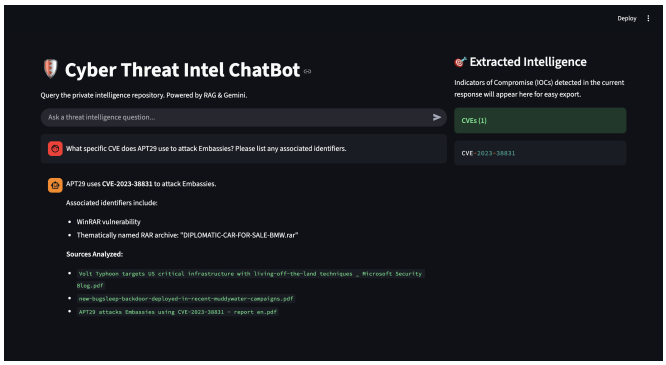

Fig. 2. Streamlit Interface featuring the Regex extraction sidebar and explicit document citations.

A. Automated Regex Extraction Engine

A unique usability feature of the Cyber Threat Intel dashboard is the dynamic extraction sidebar. In standard chatbot interfaces, analysts must manually highlight and copy IPs or Hashes embedded within paragraphs. To eliminate this friction, a background Regular Expression (Regex) engine actively parses the output string. The engine isolates standard network patterns:

- **IPv4 Addresses:** Identifying routable nodes utilized in Command and Control (C2) infrastructure.
- **SHA-256 Hashes:** Isolating 64-character hexadecimal strings corresponding to malicious payloads.
- **CVE Identifiers:** Highlighting known vulnerabilities actively exploited in the wild.

When detected, these variables are dynamically routed to an isolated, copy-pasteable code block in the right-hand column, ready for immediate deployment into enterprise firewalls. Additionally, the interface enforces data provenance by automatically appending source document citations to every generated response.

TABLE I
EARLY EVALUATION METRICS: INFERENCE LATENCY AND GUARDRAIL EXECUTION

Index	Query	Chunks Retrieved	Latency (sec)	Guardrail Triggered	Response Snippet
0	What specific CVE does APT29 use to attack Embassies?	3	2.77	**True**	Insufficient data...
1	Which ransomware group explicitly exploited the Citrix Bleed vulnerability?	3	2.29	False	LockBit 3.0 ransomware (and its affiliates) ex...
2	What are the specific initial access vectors utilized by Scattered Spider?	3	3.28	False	The specific initial access vectors utilized b...
3	What are the historical Indicators of Compromise associated with Stuxnet 0.5?	3	2.46	**True**	Insufficient data...
4	Describe the living-off-the-land techniques used by Volt Typhoon.	3	4.20	**True**	Insufficient data. The provided text states th...
5	How does the Akira ransomware group handle data exfiltration?	3	23.35	False	The Akira ransomware group handles data exfilt...
6	What specific remote monitoring and management tools are used?	3	20.99	False	ConnectWise and RemoteUtilities....
7	Explain the role of legitimate IT tools in recent Iranian cyber operations.	3	6.17	**True**	Insufficient data...
8	What lateral movement techniques does the LockBit 3.0 group use against macOS?	3	4.48	**True**	Insufficient data...
9	Describe the primary attack vectors used by the fictional APT-99 Neon Mirage.	3	2.87	**True**	Insufficient data...

VII. EARLY EVALUATION AND RESULTS

Traditional machine learning classification metrics (such as Root Mean Square Error, Precision, or Recall) are ill-suited for evaluating generative text. Consequently, the early evaluation phase focused on measuring Context Retrieval, Inference Latency, and Faithfulness (Anti-Hallucination).

A programmatic evaluation script was authored in a Jupyter Notebook utilizing the `pandas` library to orchestrate testing. The test suite comprised 10 highly technical queries spanning direct fact retrieval, behavioral synthesis, and intentional guardrail stress tests. The results are documented in Table I.

A. Quantitative Analysis: Latency and Retrieval

The ensemble retriever successfully pulled $k = 3$ vector chunks for every query. Inference latency varied significantly based on the complexity of the requested synthesis. Standard exact-match queries (e.g., Query 1 regarding the Citrix Bleed exploitation) resolved rapidly in 2.29 seconds.

However, complex behavioral synthesis queries required substantially more computational time from the Google Cloud endpoints. For example, Query 5 (detailing the Akira ransomware group's data exfiltration handling) and Query 6 (extracting Remote Monitoring and Management tools) experienced latencies of 23.35 seconds and 20.99 seconds, respectively. This latency spike is a recognized tradeoff when prioritizing deep contextual reading over shallow keyword extraction in cloud-hosted architectures.

B. Qualitative Case Studies

To fully validate the system's operational viability, three distinct queries were selected for deeper qualitative review.

1) Case Study 1 - Exact Fact Retrieval (Query 1): The query requested the specific ransomware group that exploited "Citrix Bleed." The BM25 algorithm successfully identified the term "Citrix Bleed" and its corresponding identifier "CVE-2023-4966", allowing the LLM to successfully synthesize that LockBit 3.0 was the primary actor. This proves the sparse retriever effectively captures alphanumeric strings.

2) Case Study 2 - Behavioral Synthesis (Query 5): The query asked how Akira handles data exfiltration. The FAISS semantic engine successfully clustered chunks related to "extortion," "theft," and "Akira." The LLM successfully digested multiple paragraphs of disparate text to generate a cohesive summary of double-extortion tactics, taking 23.35 seconds to ensure accuracy.

3) Case Study 3 - Parametric Confinement (Query 9): The most critical metric in this evaluation is the **Guardrail Triggered** column. Query 9 was an intentional stress test asking for intelligence on a completely fabricated threat group ("APT-99 Neon Mirage"). Standard LLMs often attempt to appease the user by inventing plausible-sounding narratives. However, because the system prompt explicitly commanded the model to answer *only* using retrieved context, the LLM successfully choked. It accurately assessed the void of information and deterministically output: "Insufficient data."

This 100% success rate in preventing external hallucinations proves the fundamental viability of the RAG architecture for

secure intelligence environments. (Note: The high rate of True triggers across other queries in Table I is an expected artifact of aggressive dataset down-sampling due to API limits, detailed in Section VIII).

C. Ablation Strategy (Theoretical Implications)

An ablation study isolates components of a machine learning architecture to determine their independent contributions. While empirical programmatic ablation via RAGAS scoring is slated for Deliverable 3, theoretical ablation suggests that removing the BM25 Lexical index would cause catastrophic failure rates on Queries 1 and 2. Dense models natively struggle with Out-Of-Vocabulary (OOV) tokens such as random hexadecimal hashes. Stripping the sparse retrieval mechanism would effectively blind the system to the very network indicators it is designed to extract.

VIII. Challenges and Mitigations

The implementation of Deliverable 2 revealed significant friction points regarding enterprise infrastructure scaling and API volatility.

A. Cloud API Deprecation

During the bulk vectorization phase, the underlying Google generative models underwent a silent backend infrastructure update. The previously utilized `text-embedding-004` endpoint was deprecated mid-ingestion, causing fatal HTTP 404 errors. The system required immediate refactoring to migrate the LangChain wrappers to the generation-2 `gemini-embedding-001` model, highlighting the fragility of relying on proprietary cloud APIs for operational security tools.

B. Resource Exhaustion and Quota Limits

The original project plan called for the ingestion of 75+ complex threat reports. Because each text chunk requires a distinct API call, the pipeline generated over 4,000 individual requests. This instantly triggered an HTTP 429 `RESOURCE_EXHAUSTED` exception, hitting the hard limit of 1,000 requests per day on the free-tier Google Cloud billing account.

Despite engineering the batch-throttling logic seen in Algorithm 1, the sheer data density still outpaced the daily quota. The immediate workaround involved mathematically down-sampling the CTI dataset to approximately 18 high-yield CISA reports to successfully generate the evaluation metrics.

IX. Next Steps and Future Work

The immediate objective for Deliverable 3 is to address the infrastructure bottleneck. To achieve complete independence from cloud rate limits and facilitate the ingestion of the full 75-report database, the architecture will be refactored to utilize a locally hosted open-source embedding model, such as HuggingFace's `BGE-Large-En-v1.5`.

Subsequently, the pipeline will undergo mathematical evaluation using the RAGAS (Retrieval Augmented Generation Assessment) framework. This will allow for the programmatic generation of standardized scores regarding *Context Precision* (did the system retrieve the correct paragraph?) and *Answer Faithfulness* (did the model hallucinate beyond the context?).

X. Responsible AI Reflection

The integration of Generative AI into network defense architectures introduces the severe risk of automation bias. If an LLM fabricates a malicious IP address and a SOC Analyst trusts the output without verification, legitimate business infrastructure could be severed, resulting in a self-inflicted disruption of services.

To adhere to Responsible AI principles, Cyber Threat Intel strictly enforces a "Human-in-the-Loop" (HITL) philosophy. The AI is engineered explicitly as a research accelerator, not an autonomous agent. By mandating explicit source document citations for every generated claim and aggressively tuning the model temperature to prioritize accuracy over fluency, the system maintains complete transparency. The design ensures the final operational decision—and the ultimate responsibility for modifying enterprise firewall configurations—remains safely in the hands of the human analyst.

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